CUSTOMER CHURN PREDICTION FOR SYRIATEL

BUSINESS PROBLEM

Customer churn is a major challenge for telecom companies because losing customers directly reduces revenue.

The goal of this project is to:

1. Predict which customers are likely to churn
2. Identify key factors driving churn
3. Provide actionable recommendations to reduce customer loss

DATA UNDERSTANDING

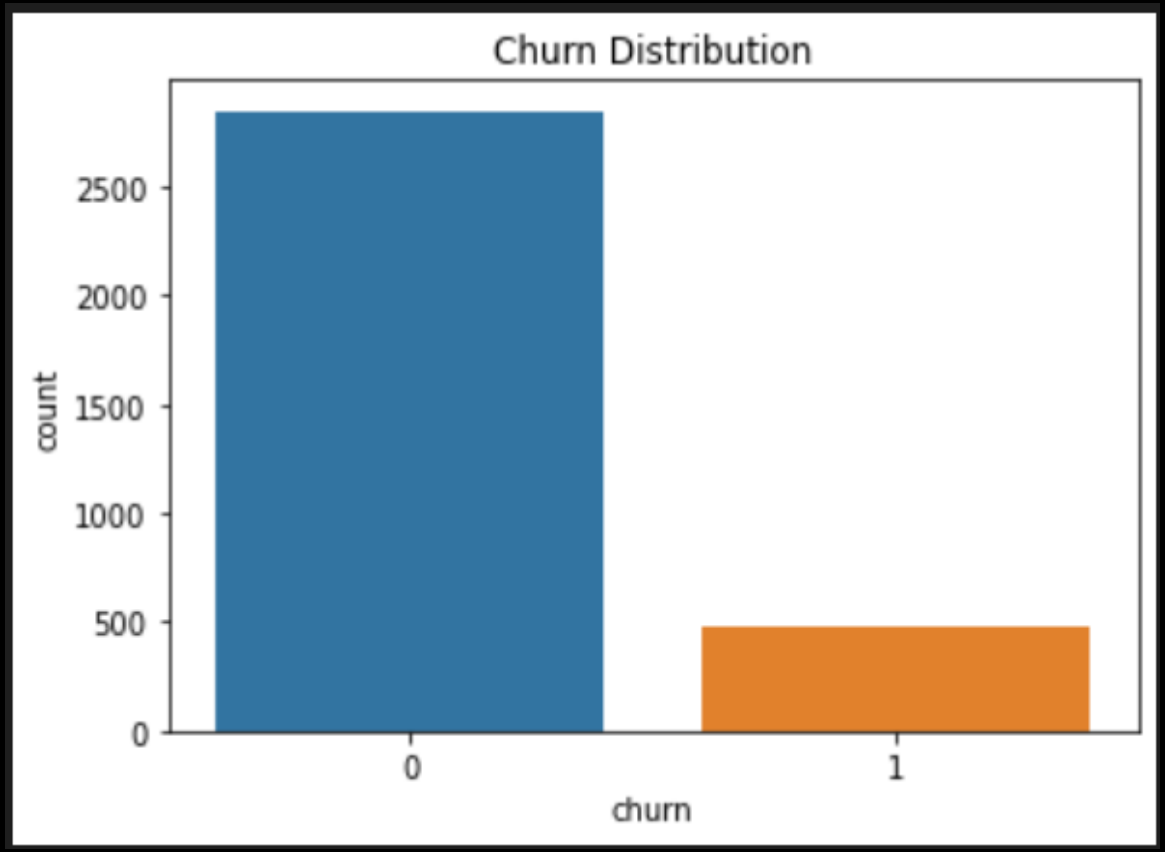
The dataset contains **3,333 customers** with **21 features**, including:

- Customer demographics (state, account length)
- Service plans (international plan, voicemail plan)
- Usage behavior (call minutes, charges)
- Customer support interactions (customer service calls)

Target variable:

- churn (1 = leave, 0 = stay)

CHURN DISTRIBUTION

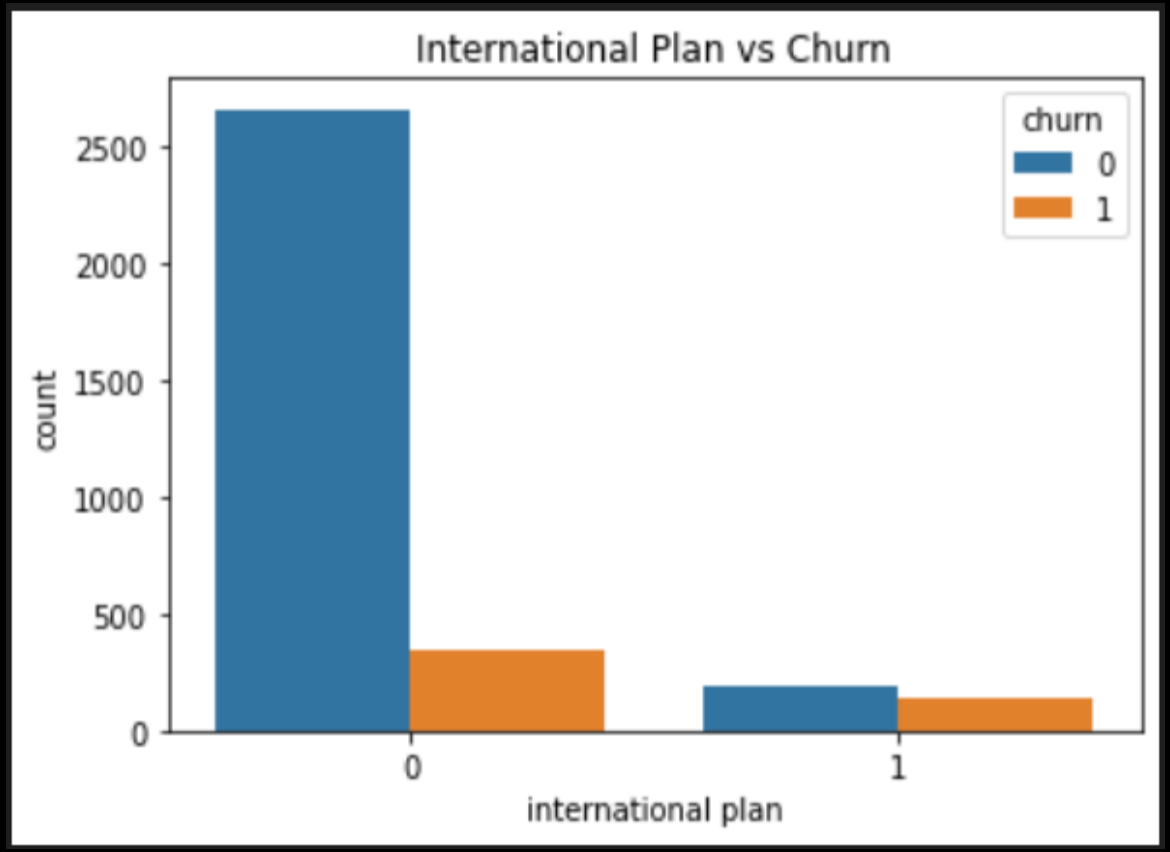


- Here we get to see there is an imbalance between the number of people churning and the ones that don't

CUSTOMER SERVICE CALLS VS CHURN

- Customers with more customer service calls are more likely to churn

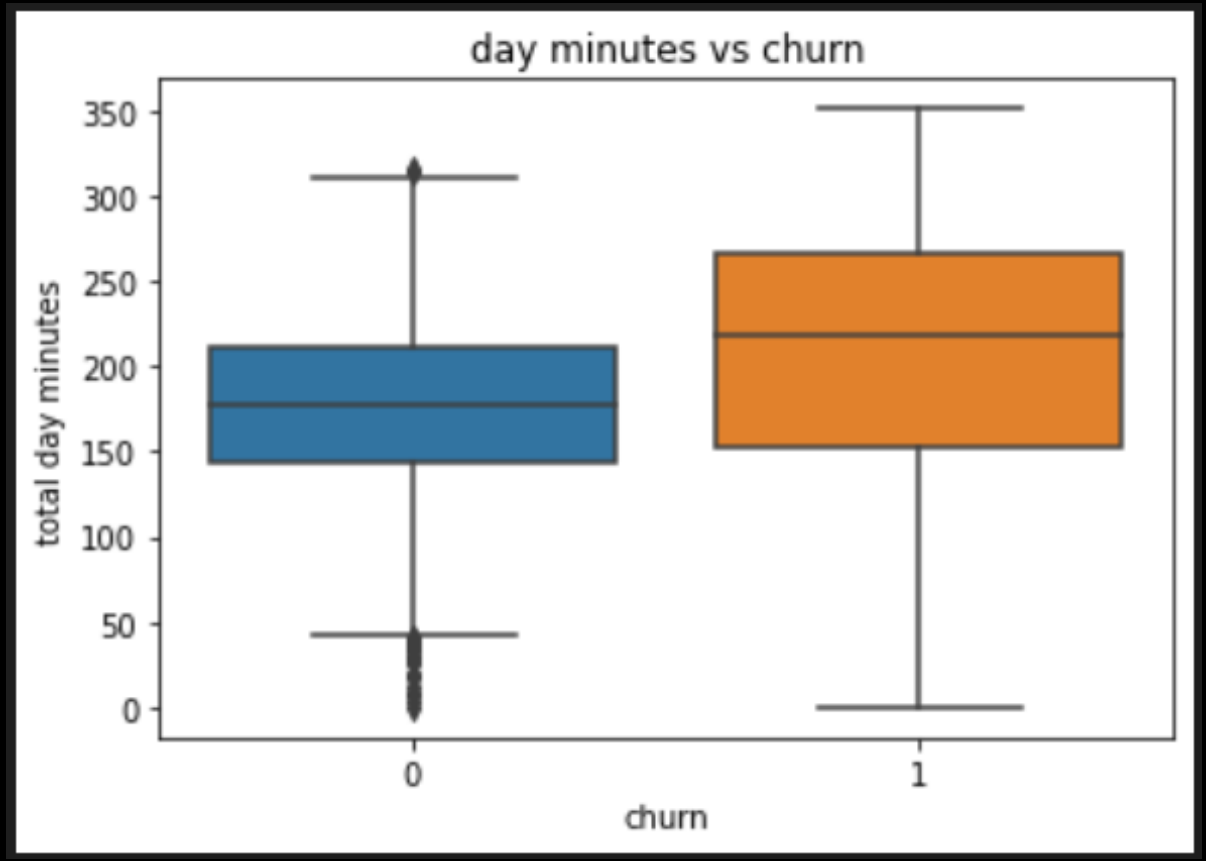
INTERNATIONAL PLAN VS CHURN



churn	0	1
international plan		
0	2664	346
1	186	137

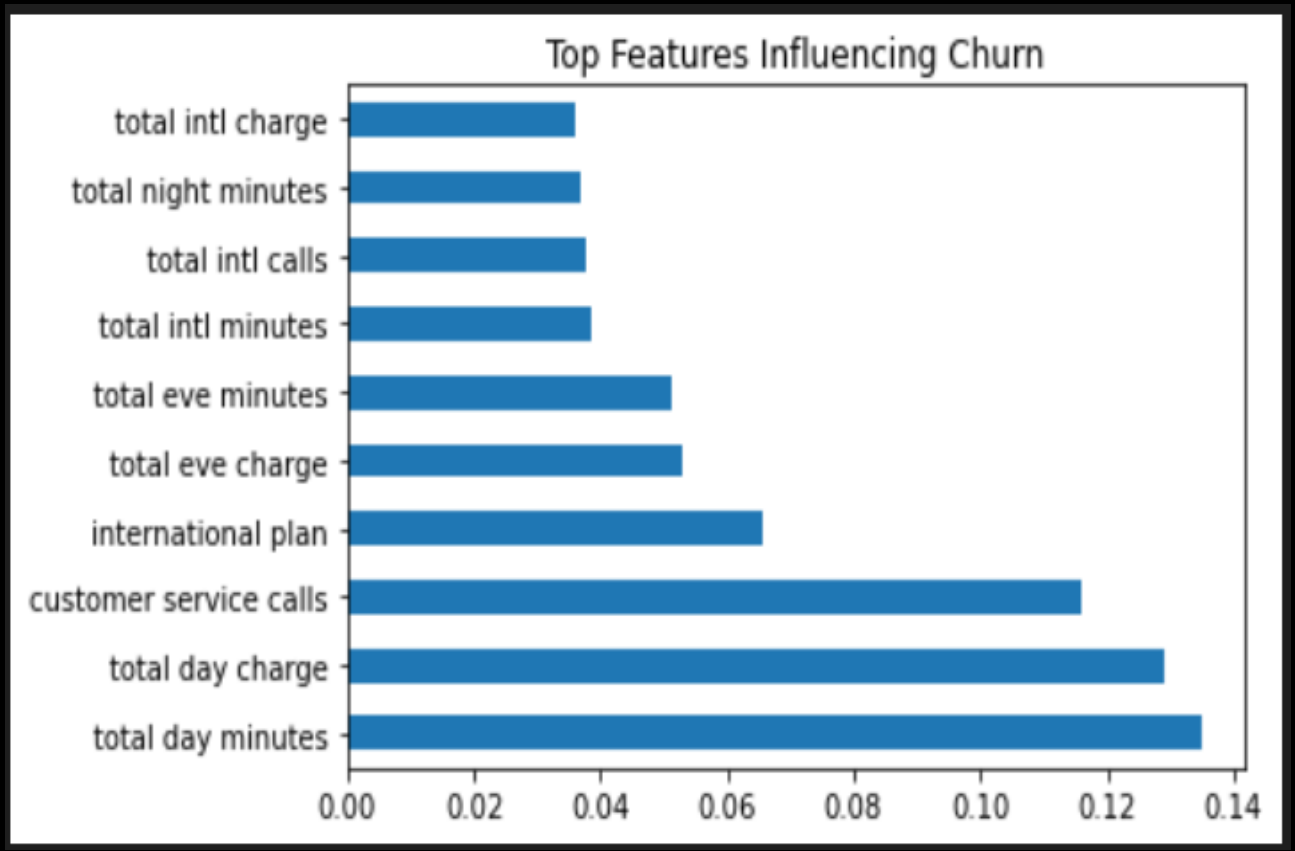
- Customers with international plans show higher churn rates

DAY MINUTES VS CHURN



- Customers who have a high daily usage and charges are linked to churn

TOP FEATURES INFLUENCING CHURN



Key Findings

- Customers with high daily usage/charges are at risk
- Customers with many customer service calls are highly likely to churn
- Customers with international plans churn more

LOGISTIC REGRESSION MODEL

```
LOGISTIC REGRESSION
Confusion Matrix:
[[  0 570]
 [  0  97]]

Classification Report:
              precision    recall  f1-score   support

           0       0.00      0.00      0.00       570
           1       0.15      1.00      0.25        97

    accuracy                           0.15       667
   macro avg       0.07      0.50      0.13       667
weighted avg       0.02      0.15      0.04       667
```

Accuracy: 15%, Precision (churn): 0.15, Recall (churn): 1.00

Completely broken model behavior:

- Predicts EVERYONE as churn
- No ability to distinguish customers

This model is useless for decision making

RANDOM FOREST MODEL

```
RANDOM FOREST
Confusion Matrix:
[[564   6]
 [ 39  58]]

Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.99      0.96       570
...
    accuracy                           0.93       667
   macro avg       0.92      0.79      0.84       667
weighted avg       0.93      0.93      0.93       667
```

Accuracy: 93%, Precision (churn): ~91%, Recall (churn): 60%

Interpretation:

- Very high accuracy
- Very high precision (few false alarms)
- Moderate recall (misses 39 churners)

This model is:

- very conservative, predicts churn only when very sure

RANDOM FOREST (AFTER TUNING)

```
[[517  53]
 [ 23  74]]
              precision    recall  f1-score   support

           0       0.96      0.91      0.93       570
           1       0.58      0.76      0.66        97

    accuracy                           0.89       667
   macro avg       0.77      0.83      0.80       667
weighted avg       0.90      0.89      0.89       667
```

Accuracy: 89%, Precision (churn): 0.58, Recall (churn): 0.76

Interpretation:

- Much higher recall (76%)
- Lower precision (more false alarms)
- Slight drop in accuracy

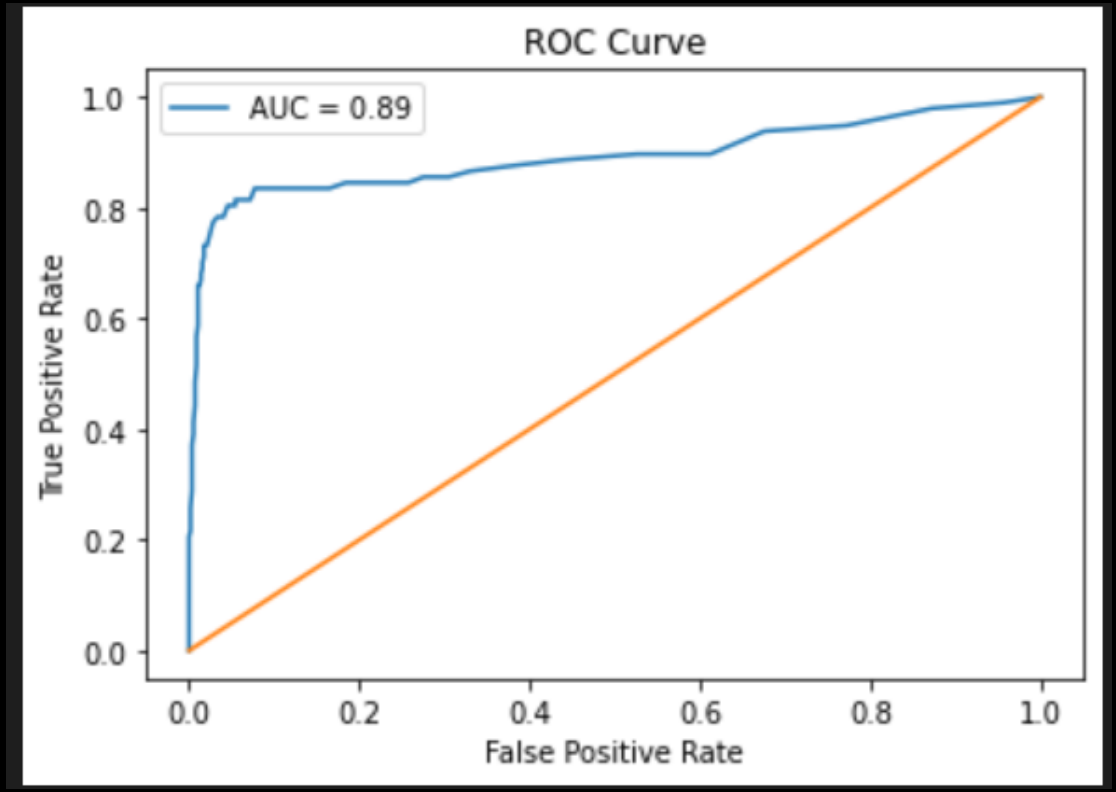
This model is:

- More aggressive and catches more churners but makes more mistakes

FINAL MODEL COMPARISON

Metric	RF (Before)	RF (After Tuning)	Logistic Regression
Accuracy	93%	89%	15%
Recall (Churn)	60%	76%	100% (bad)
Precision (Churn)	91%	58%	15%
Business Use	Good	Best	Useless

ROC CURVE AND AUC



- The ROC curve shows how well the model separates customers who will churn from those who won't.
- The AUC score tells us how good the model is overall.
- A higher AUC means better predictions.

BUSINESS RECOMMENDATIONS

- Flag high-risk customers early
- Reach out before they leave
- Faster issue resolution
- Reduce repeated complaints
- Offer discounts to heavy users
- Target users with many complaints

CONCLUSION

- The project demonstrates that machine learning can effectively predict customer churn and provide actionable insights to improve customer retention.
- By acting on these insights, SyriaTel can reduce churn and improve customer retention.